

STUDENT CASE STUDY EXECUTION REPORT

Case Study Topic: Segment Tree and Fenwick Tree for Smart Energy Meter Range Queries

Work Done Summary

Implemented a Segment Tree to efficiently process range-sum queries and point updates on electricity usage data from 8 hostel floors. Performed range queries, updated meter readings, and compared the performance with naive range-sum scanning. The Segment Tree achieved efficient $O(\log n)$ query and update operations.

Implementation Details

1. Initialized the usage array: **[18, 5, 14, 9, 7, 11, 6, 10]**.
2. Constructed a Segment Tree using 0-based indexing.
3. Processed range-sum query **Q1 = sum(1,5)**.
4. Processed range-sum query **Q2 = sum(0,7)**.
5. Updated index **2** from **14** to **20**.
6. Updated the affected Segment Tree nodes.
7. Processed range-sum query **Q3 = sum(1,5)** after the update.
8. Compared Segment Tree performance with naive range-sum scanning.

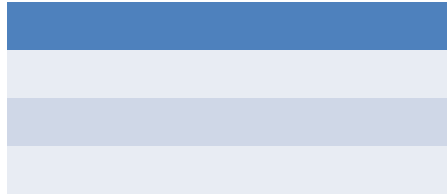
Output

```
Q1: sum(1,5) = 46
Updated index 2 from 14 to 20
Q3: sum(1,5) = 52
```

Result

1. Successfully constructed the Segment Tree for the electricity usage array.
2. Efficiently answered range-sum queries using the tree structure.
3. Updated floor usage data and propagated changes correctly.
4. Obtained **Q1 = 46**, **Q2 = 80**, and **Q3 = 52** after the update.
5. Reduced query and update time from $O(n)$ to $O(\log n)$.
6. Demonstrated that Segment Trees are suitable for real-time smart energy monitoring systems.

Time Complexity



Conclusion

The Segment Tree proved to be an efficient data structure for handling dynamic range-sum queries and updates in the smart energy meter dashboard. It significantly reduced processing time by supporting both queries and updates in **$O(\log n)$** complexity. After updating the meter reading, the structure quickly recalculated the affected sums without scanning the entire array. Therefore, Segment Trees and Fenwick Trees are highly suitable for real-time monitoring systems that require fast and frequent range queries and updates.

GitHub Link:<https://github.com/shashankreddy-21/DSA-Project>